

UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy
DEGREE OF M.Sci. WITH HONOURS
FOURTH-YEAR FINAL EXAMINATION

03 01087

LM NUCLEAR PHYSICS

SEMESTER 2 EXAMINATIONS 2022/23

Time Allowed: 1 hour 30 minutes

Answer one question from Section 1 and one question from Section 2.

Section 1 consists of two questions and carries 50% of the marks for the examination. Answer **one** question from this Section. If you answer more than one question, credit will only be given for the best answer.

Section 2 consists of a single question and carries 50% of the marks for the examination.

The approximate allocation of marks to each part of a question is shown in brackets [].

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

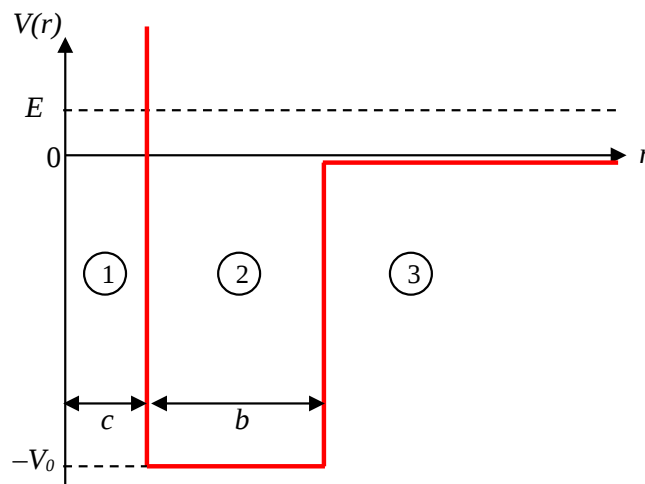
A formula sheet and a table of physical constants and units that may be required will be found at the end of this question paper.

SECTION 1

Answer **one** question from this Section. If you answer more than one question, credit will only be given for the best answer.

1. Nucleon-nucleon scattering can be studied on the basis of the approximate potential shown in the figure below. The solid line is the approximate potential for which there are three regions:

region (1) of width c (where the potential is infinite),
 region (2) of width, b , where the potential has depth V_0 , and
 region (3) where the potential is equal to 0.



- (a) Solve the Schrödinger equation for s-wave scattering and write down the general solutions for the wave functions for each of the three regions. [9]

- (b) Show that the phase shift, δ_0 , for s-wave scattering is given by the condition:

$$k_2 \cot(k_2 b) = k_3 \cot[k_3(c + b) + \delta_0],$$

where $k_2 = \sqrt{2m(V_0 + E_{cm})}/\hbar$ and $k_3 = \sqrt{2mE_{cm}}/\hbar$. [5]

- (c) Determine the parameter c using the following experimental data:

$$E_{lab} = 350 \text{ MeV}, \delta_0 = -0.25, V_0 = 73 \text{ MeV and } b = 1.337 \text{ fm.}$$

[You may assume a nucleon mass of $1.67 \times 10^{-27} \text{ kg}$.] [8]

- (d) Determine the cross section for neutron scattering on a proton at energy $E_{cm} = 1 \text{ MeV}$ using the following experimentally determined parameters $V_0 = 73 \text{ MeV}$, $b = 1.337 \text{ fm}$, $c = 0.4 \text{ fm}$.

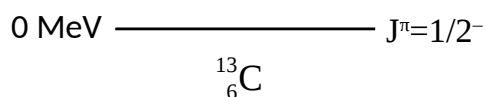
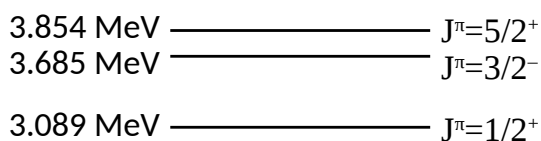
Demonstrate that the assumption of s-wave scattering is valid for $E_{cm} = 1 \text{ MeV}$.

[Hint: You may use $\sigma_{tot} = \frac{4\pi}{k^2} \sin^2 \delta_0$ and assume the neutron and proton masses are equal.]

[8]

ANY CALCULATOR

2. (a) The spectrum of low-lying states of $^{13}_6\text{C}$ is shown in the figure below.



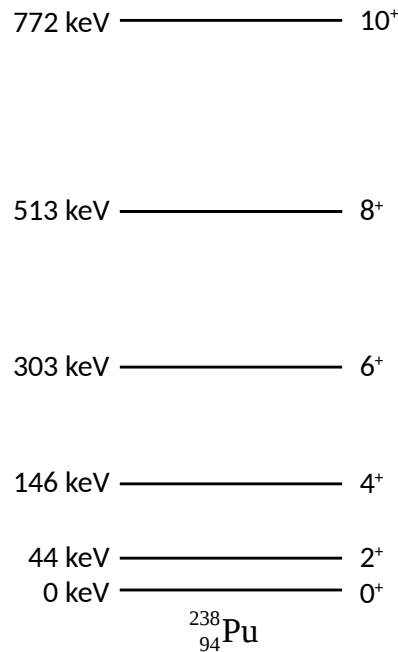
- i. Explain the shell model configuration of the four states in $^{13}_6\text{C}$ shown. [Hint: The first six shell model states are: $1s_{\frac{1}{2}}$, $1p_{\frac{3}{2}}$, $1p_{\frac{1}{2}}$, $1d_{\frac{5}{2}}$, $2s_{\frac{1}{2}}$, and $1d_{\frac{3}{2}}$.] [9]
 - ii. Give reasons why the next excited states above 3.854 MeV appear at excitation energies around 7 MeV. [3]
 - iii. Write down and explain the shell model configuration for the ground state and first excited state of $^{14}_7\text{N}$. [4]
- (b)
- i. Explain why some ground states of even-even actinide nuclei can decay by the emission of alpha particles with several different energies. Explain all the effects that lead to the lower energy alpha particles having lower intensities. [4]
 - ii. Two isotopes of thorium, $^{232}_{90}\text{Th}$ ($t_{1/2} = 1.4 \times 10^{10}$ years) and $^{218}_{90}\text{Th}$, undergo alpha decay. A third thorium isotope, $^{228}_{90}\text{Th}$, has a half-life of 1.912 years with $E_\alpha = 5.423$ MeV. Using the Geiger-Nuttall rule (which can be obtained from expressions given in the formula sheet), calculate the half-life ratio, $t_{1/2}(^{232}\text{Th})/t_{1/2}(^{218}\text{Th})$.
 [You may find the follow data useful: $\Delta m(^{232}_{90}\text{Th}) = 35446.71094$ keV, $\Delta m(^{228}_{88}\text{Ra}) = 28940.19336$ keV, $\Delta m(^{218}_{90}\text{Th}) = 12366.74707$ keV, $\Delta m(^{214}_{88}\text{Ra}) = 92.74000$ keV and $\Delta m(^4_2\text{He}) = 2424.91577$ keV.] [10]

SECTION 2

Answer **one** question from this Section.

3. (a) Explain why quantum mechanics forbids the rotation of a sphere. [4]

- (b) The ground-state band of $^{238}_{94}\text{Pu}$ is shown in the figure below.



Use the level scheme above to calculate the average moment of inertia for the band and explain the nature of the states [7]

- (c) The quadrupole moment of a deformed nucleus is given by

$$Q = \frac{3}{\sqrt{5\pi}} R_{av}^2 Z \beta (1 + 0.16\beta),$$

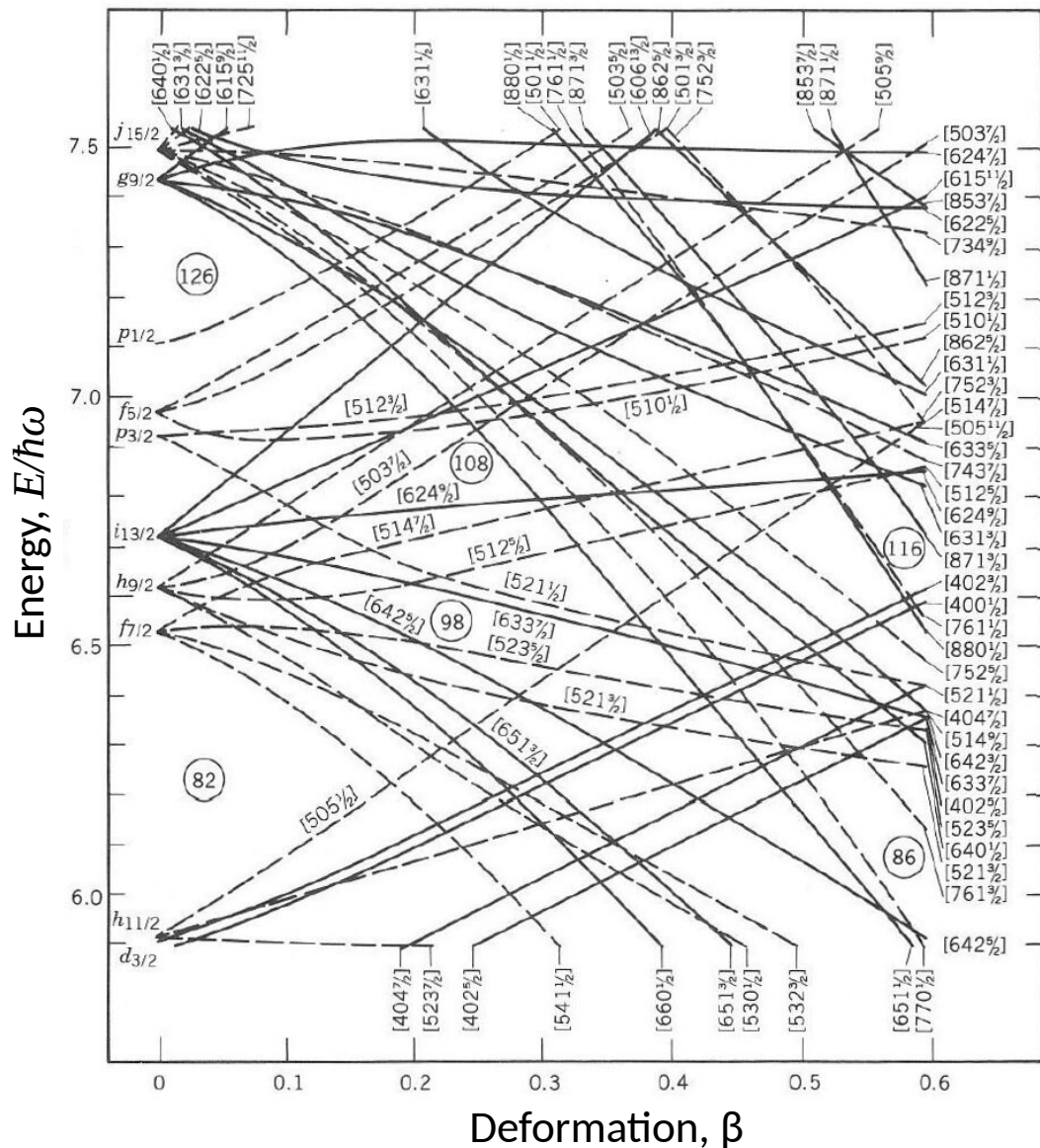
where $\beta = \beta_2 = \frac{4}{3} \sqrt{\frac{\pi}{5}} \frac{\Delta R}{R_{av}}$ and $R_{av} = 1.2A^{1/3}$ fm.

Define ΔR and explain why the deformation parameter, β , is often written as β_2 . To what shapes does β pertain? [4]

- (d) For $^{238}_{94}\text{Pu}$ $\Delta R = 1.655$ fm. Calculate the corresponding quadrupole moment in barns. [4]

Question continues on next page.

(e) Below is part of the Nilsson diagram for heavy nuclei.



The orbital labels in the square brackets are $[N N_z \Lambda \Omega]$. Define each of these parameters for a prolate deformed nucleus and state how the parity of the orbitals can be obtained. [5]

(f) Using this Nilsson diagram write down the Nilsson orbitals populated by the last neutron and last proton in ${}^{238}_{94}\text{Pu}$. [6]

Useful Formulae for Nuclear Physics

Coulomb barrier, $V_{Coul} = \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 R}$, where $R = 1.2A^{1/3}$ fm

Radial Schrödinger equation, $-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[V(r) + \frac{l(l+1)\hbar^2}{2mr^2} \right] u(r) = Eu(r)$,

where $R(r) = \frac{u(r)}{r}$

Deuteron wave function matching condition, $k_1 \cot(k_1 R) = -k_2$

where $k_1 = \sqrt{\frac{2m(E + V_0)}{\hbar^2}}$ and $k_2 = \sqrt{-\frac{2mE}{\hbar^2}}$

Magnetic moment, $\langle \mu \rangle =$

$$\frac{1}{2(j+1)} \{ g_l [j(j+1) + l(l+1) - s(s+1)] + g_s [j(j+1) + s(s+1) - l(l+1)] \} \mu_N$$

Alpha-decay half-life, $t_{1/2} = \ln(2) \frac{R}{c} \sqrt{\frac{mc^2}{2(V_0 + Q)}} \exp \{2G\}$,

where G is the Gamow factor, $G = \sqrt{\frac{2m}{\hbar^2 Q}} \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0} \left[\frac{\pi}{2} - 2 \left(\frac{Q}{B} \right)^{1/2} \right]$

Beta-decay ft values, $f(Z, T_e) t_{1/2} = \ln(2) \frac{2\pi^3 \hbar^7}{g_F^2 |M_{fi}|^2 m_e^5 c^4}$

Internal conversion coefficients, $\alpha(ML) \approx \frac{Z^3}{n^3} \left(\frac{e^2}{4\pi\epsilon_0 \hbar c} \right)^4 \left(\frac{2m_e c^2}{E} \right)^{L+3/2}$ and

$$\alpha(EL) \approx \frac{Z^3}{n^3} \left(\frac{L}{L+1} \right) \left(\frac{e^2}{4\pi\epsilon_0 \hbar c} \right)^4 \left(\frac{2m_e c^2}{E} \right)^{L+5/2}$$

Weisskopf estimates (for E_γ in MeV),

$$\lambda(E1) = 1.0 \times 10^{14} A^{2/3} E_\gamma^3$$

$$\lambda(E4) = 1.1 \times 10^{-5} A^{8/3} E_\gamma^9$$

$$\lambda(M1) = 3.1 \times 10^{13} E_\gamma^3$$

$$\lambda(M4) = 3.3 \times 10^{-6} A^2 E_\gamma^9$$

$$\lambda(E2) = 7.4 \times 10^7 A^{4/3} E_\gamma^5$$

$$\lambda(E5) = 2.4 \times 10^{-12} A^{10/3} E_\gamma^{11}$$

$$\lambda(M2) = 2.2 \times 10^7 A^{2/3} E_\gamma^5$$

$$\lambda(M5) = 7.4 \times 10^{-13} A^{8/3} E_\gamma^{11}$$

$$\lambda(E3) = 3.5 \times 10^1 A^2 E_\gamma^7$$

$$\lambda(M3) = 1.1 \times 10^1 A^{4/3} E_\gamma^7$$

Physical Constants and Units

Acceleration due to gravity	g	9.81 m s^{-2}
Gravitational constant	G	$6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Ice point	T_{ice}	273.15 K
Avogadro constant	N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
[N.B. 1 mole $\equiv$ 1 gram-molecule]		
Gas constant	R	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23} \text{ J K}^{-1} \equiv 8.62 \times 10^{-5} \text{ eV K}^{-1}$
Stefan constant	σ	$5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Rydberg constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
	$R_\infty hc$	13.606 eV
Planck constant	h	$6.626 \times 10^{-34} \text{ J s} \equiv 4.136 \times 10^{-15} \text{ eV s}$
	$h/2\pi$	$\hbar$ $1.055 \times 10^{-34} \text{ J s} \equiv 6.582 \times 10^{-16} \text{ eV s}$
Speed of light <i>in vacuo</i>	c	$2.998 \times 10^8 \text{ m s}^{-1}$
	$\hbar c$	197.3 MeV fm
Charge of proton	e	$1.602 \times 10^{-19} \text{ C}$
Mass of electron	m_e	$9.109 \times 10^{-31} \text{ kg}$
Rest energy of electron		0.511 MeV
Mass of proton	m_p	$1.673 \times 10^{-27} \text{ kg}$
Rest energy of proton		938.3 MeV
One atomic mass unit	u	$1.66 \times 10^{-27} \text{ kg}$
Atomic mass unit energy equivalent		931.5 MeV
Electric constant	ϵ_0	$8.854 \times 10^{-12} \text{ F m}^{-1}$
Magnetic constant	μ_0	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Bohr magneton	μ_B	$9.274 \times 10^{-24} \text{ A m}^2 (\text{J T}^{-1})$
Nuclear magneton	μ_N	$5.051 \times 10^{-27} \text{ A m}^2 (\text{J T}^{-1})$
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	$7.297 \times 10^{-3} = 1/137.0$
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12} \text{ m}$
Bohr radius	a_0	$5.2918 \times 10^{-11} \text{ m}$
angstrom	$\text{\AA}$	10^{-10} m
barn	b	10^{-28} m^2
torr (mm Hg at 0 °C)	torr	$133.32 \text{ Pa (N m}^{-2})$

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches must be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.